

From Acceleration to Feasibility: HPC–AI Convergence, Energy Constraints, and Public Policy for Technological Sovereignty in Colombia

Esteban Hernández

CyberColombia, Bogotá, Colombia
`esteban.hernandez@cybercolombia.org`

Abstract

The convergence between high-performance computing (HPC) and artificial intelligence (AI) is reshaping scientific computing methodologically, organizationally, and in its infrastructure. AI no longer merely consumes accelerators: it operates as a component of simulation chains, as a surrogate for simulation via inference, and in optimization, data analysis, and code generation. This shift broadens computational exploration but introduces problems of validation, reproducibility, and technological dependence. This article analyzes that tension from the Colombian case. Drawing on regional and national HPC infrastructure reports, on the recent portfolio of national science-and-technology funding instruments for AI and quantum technologies, on the National AI Policy, on the current legislative landscape, and, for calibration, on the most recent TOP500 list and public information on hyper-scale datacenters, it examines three questions: what is the real baseline of the Colombian HPC ecosystem? what economic, logistical, and energy implications would supporting high-quality foundation-model workloads entail? and how does the design of public instruments strengthen or weaken technological sovereignty? The analysis introduces an empirical calibration of three power regimes —the sub-megawatt regime of Latin American public science, the tens of megawatts of frontier scientific supercomputers, and the hundreds to thousands of megawatts of frontier AI training campuses— and shows that Colombian instruments lack precisely that calibration. The central argument: sovereignty in AI is not rhetorical self-sufficiency, but the operational capacity to validate models, host critical workloads, protect sensitive data, and sustain advanced infrastructure under long-term scientific criteria.

Keywords: HPC, artificial intelligence, scientific computing, datacenters, energy, public policy, CONPES 4144, technological sovereignty, Colombia, Latin America

1 Introduction

The current phase of interaction between HPC and AI differs from the stage in which machine learning depended on accelerators designed for traditional scientific workloads. Today the integration is methodological: AI models participate in simulation workflows, reduce dimensionality, build inference-based surrogates of simulation, assist programming, and accelerate the analysis of massive data. This opens opportunities, but it shifts the central problem from hardware availability toward the epistemological and operational quality of results, because acceleration does not equal validity. HPC rests on numerical analysis, uncertainty quantification, reproducibility, comparison against references, and benchmarking; many AI systems operate as high-capacity statistical approximators whose empirical performance guarantees neither physical correctness, nor out-of-distribution stability, nor sufficient traceability. In countries with limited infrastructure the gap is amplified: it is possible to consume advanced AI services before having the local capacity to audit them.

In Colombia the problem is simultaneously scientific, infrastructural, and one of public policy: the task is to define under which material and institutional conditions the adoption of AI can be sustained with scientific rigor and strategic autonomy. We examine that relationship in light of three recent instruments: SGR Call 46 [6], Call No. 966 of 2025 [7], and Call No. 976 of 2026, Colombia Inteligente [8]. The analysis is calibrated against two external references: the June 2026 TOP500 list [16], which reports the measured power consumption of the world’s largest supercomputers, and the public technical information on hyperscale AI training campuses operated by Meta, xAI, and Google [14, 15, 12].

2 Methodology

This work is a comparative documentary analysis. The evidence base is a set of public and technical sources read under a single analytical framework rather than aggregated statistically, organized into three groups. The first group establishes the Colombian and regional baseline: the third infrastructure report of the SCALAC HPC Observatory [1] and the report *Status of HPC in Colombia 2026* [4], which characterizes the institutional base, funding models, and gaps in training and infrastructure. The second group gathers the public-policy cases: the three instruments of the Ministry of Science together with the definitive bank of eligible and fundable projects of Call 966 of 2025 [6, 7, 8, 9], CONPES 4144 [20], and the 2024–2025 legislative record [21, 29, 30]. The third group calibrates the material scale: the June 2026 TOP500 list, from which the measured power consumption of reporting systems is taken [16], public information on hyperscale AI campuses [14, 15, 12, 13], and official documentation from Latin American national centers [22, 23, 24]. When a system does not report power, it is excluded from the corresponding statistic rather than imputed.

The national survey underlying [4] consisted of a questionnaire applied to eleven institutional HPC capacities distributed across seven cities (Bogotá, Medellín, Cali, Manizales, Bucaramanga, Cartagena, and Barranquilla), five of them in Bogotá, conducted between March and October 2025, with an instrument covering installed infrastructure, funding models, staffing, training, and operational needs. The eleven institutional responses analyzed here, together with the survey instrument, are available as an open dataset [2]; a DOI-issuing deposit in Zenodo is planned to supersede the current institutional URL. Since the author is affiliated with the organization that produced that survey, the declaration at the end of the article details the mitigation measures adopted, chiefly systematic triangulation with independent sources.

The analysis is organized along three axes: installed capacity, infrastructure requirements, and coherence between normative design and material feasibility.

3 HPC–AI Convergence and the Validation Problem

The friction between speed and verification arises at several levels. Inference-based surrogates of simulation and emulators drastically reduce cost, but they can hide errors outside the training domain. Automatic code generation accelerates implementation, yet may introduce silent failures without rigorous review protocols. Large pretrained models concentrate methodological assumptions and design decisions in external providers, reducing local capacity to inspect data, architecture, and evaluation criteria.

Thus, the pertinent question is not whether AI should replace classical HPC, but how to integrate it into a verification architecture: AI accelerates exploration, synthesis, and prioritization; HPC remains the reference layer for reproducibility, numerical contrast, and audit. The convergence is not substitution, but a reordering of the scientific stack.

Table 1: Latin American systems in the TOP500, June 2026, with sustained performance ($R_{\max}$, PFlop/s) and, where disclosed, power consumption. "n/r" = power not reported. Only Santos Dumont and Clementina XXI are open public science or state infrastructure; IARA belongs to a private R&D institute; the rest is private industry. Source: official TOP500 data file, June 2026 [16].

Rank	System	Site (country)	Segment	$R_{\max}$ (PFlop/s)	Power (MW)
37	Harpia	Petrobras (BR)	Industry	75.2	2.01
117	Pégaso	Petrobras (BR)	Industry	19.1	1.03
141	Santos Dumont	LNCC (BR)	Academic	14.3	0.31
206	Dragão	Petrobras (BR)	Industry	9.0	0.94
248	Gaia	Petrobras (BR)	Industry	7.0	0.57
289	Clementina XXI	SMN (AR)	Public research	5.4	n/r
324	Atlas	Petrobras (BR)	Industry	4.4	0.55
362	Gemini	Petrobras (BR)	Industry	3.9	0.29
369	IARA	SiDi (BR)	Private R&D	3.7	n/r
380	NOBZ1	MBZ (BR)	Industry	3.5	n/r
408	Fênix	Petrobras (BR)	Industry	3.2	0.39

4 Regional and National Baseline

The third infrastructure report of the SCALAC HPC Observatory [1], together with the findings of the Euro–Latin American RISC2 network [25], offers a relevant regional starting point by highlighting dimensions that are key to the AI discussion: workload orientation, connectivity, energy provision, effective utilization, and operating conditions. Its value lies not only in comparing theoretical performance, but in incorporating elements ignored in public debates — staffing levels, operational heterogeneity, academic connectivity, actual utilization, and fragility of sustainment—, an approach that is pertinent for Colombia, where having equipment does not exhaust the capacity problem. The diagnosis shows capacities concentrated in a few countries, with clear leadership from Brazil and relevant presence of Mexico, Argentina, and Chile; for Colombia this implies a less consolidated position in large-scale public infrastructure, a reading consistent with the Latin American Artificial Intelligence Index [18].

The 2026 CyberColombia report [4], based on the national survey described in Section 2, shows an active ecosystem, with public and private centers, participation in academic networks, and presence in venues such as the HPC Summer School and CARLA, coordinated in part through RedCLARA and RENATA [19]. But it also reveals structural limits: mixed or self-generated funding, weak stable long-term financing, unresolved needs in training and infrastructure, and insufficient articulation with public policy. Colombia does not lack an HPC community: it has a significant but dispersed one, with talent, operational experience, and a culture of collaboration, yet without the national coordination, sustained funding, or network integration required for cumulative sovereign capacity.

4.1 The Regional Scale in Numbers: Latin America in the TOP500

To calibrate the diagnosis with measured data, we examined the June 2026 TOP500 list [16] for Latin American systems. The result is stark: only eleven of the 500 are in the region —ten in Brazil and one in Argentina— and none in Mexico, Chile, or Colombia. Seven belong to a single private operator, the oil company Petrobras, and two more to private software and R&D companies (MBZ and SiDi); only two are open public science or state infrastructure (Santos Dumont at the LNCC and Clementina XXI at the Argentine national weather service), so the footprint of public science is even thinner than the raw count. Table 1 lists the eleven systems with their sustained performance and, where reported, their power consumption.

Two emblematic national centers do not even reach the entry threshold: Chile’s NLHPC operates Guacolda-Leftrarú Epu (about 9,956 cores, on the order of 0.77 PFlop/s in double

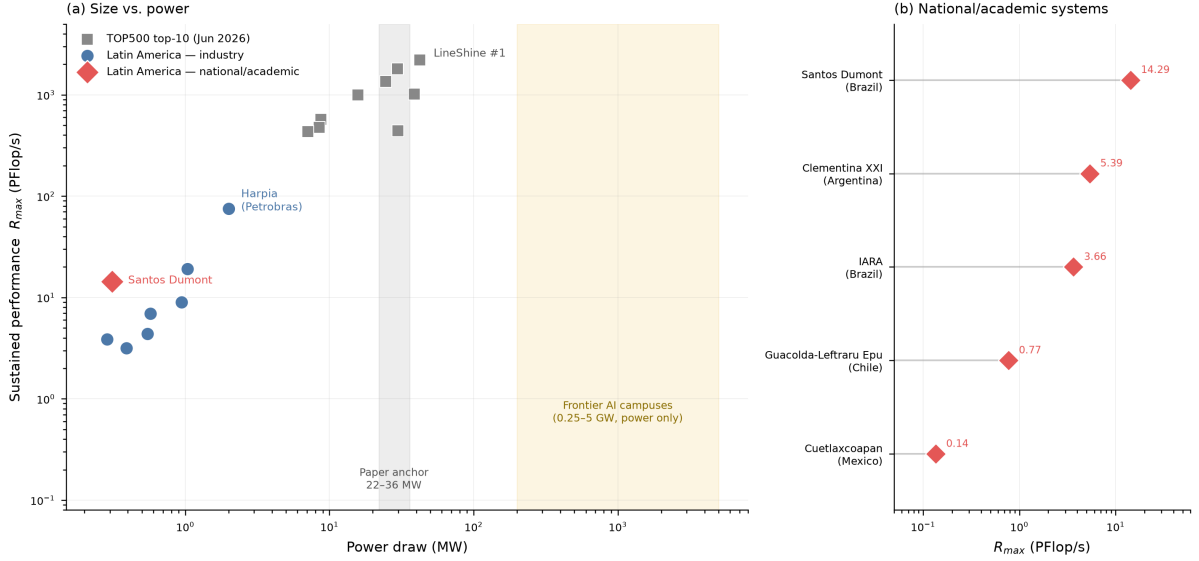


Figure 1: Latin American HPC on a common size–power plane. (a) Sustained performance versus power consumption (log–log): Latin American systems (industry in blue, public science in red) sit one to three orders of magnitude below the TOP500 top ten (gray) in performance, and one to two orders of magnitude below in power; shown for reference are the 22–36 MW band of frontier scientific supercomputers and the 0.25–5 GW frontier AI band. (b) Public-science systems by sustained performance, including the sub-petascale flagships of Chile (NLHPC) and Mexico (LNS) that do not enter the TOP500. Sources: TOP500 June 2026 [16] and official center documentation [22, 23, 24].

precision) [22], and the LNS at BUAP in Mexico operates Cuertlaxcoapan (about 0.14 PFlop/s) [23]; both are sub-petascale, an order of magnitude or more below TOP500 entry. Brazil’s Santos Dumont, the region’s main open-science machine, was upgraded in 2025 to a peak near 18.85 PFlop/s [24]: even the leader sits two orders of magnitude below the exascale frontier.

These measurements define a third power regime that the frontier anchor of Section 6 does not capture on its own. The tens-of-megawatts band (22–36 MW) describes the world’s frontier scientific supercomputers and the gigawatt band the frontier AI campuses, but the actual Latin American public-science systems operate around 0.1–2 MW: Santos Dumont delivers 14.3 PFlop/s with just 0.31 MW, and the largest regional system, Harpia, consumes about 2 MW. Figure 1 places them against the TOP500 top ten and the frontier AI campuses on a common size–power plane. A credible first-generation Colombian national system therefore belongs to this regime —units-to-tens of petaflops at roughly one megawatt, a Santos Dumont-scale objective, not the frontier. Section 6 draws out what that calibration implies for public policy.

5 What Building Infrastructure for Foundation Models Entails

The expression "building a datacenter to run foundation models" is often used imprecisely. Three scenarios should be distinguished: local inference on large open models; fine-tuning and sectoral adaptation of advanced models; and frontier training or co-training. Each implies different orders of magnitude in investment, energy, networking, storage, support, and exposure to restricted supply chains.

Public information on the infrastructures of Meta [10, 14], NVIDIA [11], and Google [12] shows that frontier AI depends on datacenter-scale engineering: buying accelerators is not

enough. The functional unit is an integrated system of compute, interconnect, storage, power, cooling, security, monitoring, and operations, characteristic of strategic infrastructure.

For Colombia it is methodologically sounder to differentiate a national advanced computing center oriented to reproducible science, sensitive inference, simulation, and model adaptation, from a frontier training infrastructure. The former is difficult but plausible—and, as Section 4.1 shows, its realistic entry scale is that of Latin American public-science systems—; the latter requires a much larger financial, energy, and logistical scale.

6 Economic, Logistical, and Energy Constraints

The economics of frontier AI is determined by capital expenditure, technology refresh, support contracts, and recurring operations. As Section 5 stressed, the relevant unit is the integrated campus rather than the accelerator; its cost is dominated by network switching, power and cooling plant, backup, physical security, and specialized staff, not by the servers alone. Logistics is just as severe. Advanced accelerators depend on tight supply chains, long procurement times, and at times geopolitical sensitivity, and they add site needs: technical space, maintenance access, low-latency connectivity, fire protection, spare parts, and staff experienced in continuous operations. Often the bottleneck is not buying the compute, but the maturity of the site and the organization hosting it.

The critical variable is energy. AI-intensive workloads turn computational ambition into continuous electrical demand and, therefore, into a problem of power, cooling, and continuity [17]. Even intermediate deployments quickly move from laboratory scale to multi-megawatt operations, a threshold beyond which planning must incorporate power quality, redundancy, heat rejection, water management, environmental permits, and exposure to electricity prices.

These constraints are not abstract in the Colombian case. The National Interconnected System has an effective installed capacity on the order of 20 GW, with a hydro share close to two-thirds of generation [26], which exposes supply to hydroclimatic variability: the 2023–2024 El Niño drove reservoirs to critical levels and brought the system close to rationing, and well-known delays persist in transmission projects and in the incorporation of new generation identified in UPME planning [27]. In that context, a continuous load on the order of one megawatt—the regime of Section 4.1—is manageable within existing urban infrastructure; a load of tens of megawatts, by contrast, competes with the demand of intermediate cities and requires joint planning with the system operator and UPME from project design onward, not as a subsequent formality. Any serious national AI infrastructure project must be designed together with the power system, not as a late extension of it: a hydro-heavy matrix can be an advantage, but it does not substitute for analysis of transmission, resilience, climate risk, water security, operational continuity, or substation siting.

CyberColombia’s *Introduction to Supercomputing* presentation [5] translates this to an intuitive scale: it distinguishes prototypes that fit on workstations or GPU nodes from the training of LLM-type models, which can require hundreds to thousands of GPUs, high-performance storage, and supercomputing infrastructure. The 22–36 MW band of these frontier solutions is easier to grasp in national terms: at Colombia’s average electricity consumption of about 1,600 kWh per inhabitant per year—a continuous draw of roughly 183 W per person—a load of 22–36 MW equals the year-round electricity use of some 120,000 to 200,000 Colombians, the population of an intermediate Colombian city [28]. An AI factory pressures not only the budget, but the power grid, territorial planning, the cooling strategy, and the political legitimacy of allocating significant energy capacity.

Measured data confirm that band as a descriptor of the world scientific frontier. In the June 2026 TOP500 list, the five largest scientific supercomputers that report power sit between 24.6 and 42.2 MW: Frontier at 24.6 MW, El Capitan at 29.7 MW, Fugaku at 29.9 MW, Aurora at 38.7 MW, and LineShine, first on the list, at 42.2 MW [16] (Table 2). Three fall within the

Table 2: Top ten systems of the June 2026 TOP500 list, with sustained performance and, where disclosed, power consumption and energy efficiency. Not all systems report power (e.g., Eagle). Source: official TOP500 data file, June 2026 [16].

Rank	System	Country	PFlop/s	Power (MW)	GFlop/W
1	LineShine	China	2198	42.2	52.1
2	El Capitan	United States	1809	29.7	60.9
3	Frontier	United States	1353	24.6	55.0
4	Aurora	United States	1012	38.7	26.2
5	JUPITER Booster	Germany	1000	15.8	63.3
6	HPC7	Italy	572	8.7	65.4
7	Eagle	United States	561	—	—
8	HPC6	Italy	478	8.5	56.5
9	Fugaku	Japan	442	29.9	14.8
10	Alps	Switzerland	435	7.1	61.0

22–36 MW range, validating that band as an envelope of the world scientific frontier; since only 29 of the top 50 systems report power, this baseline understates the fleet’s real demand.

The frontier of AI infrastructure, however, already sits one to two orders of magnitude above the scientific frontier. xAI’s Colossus 1 cluster in Memphis reportedly demanded some 250 MW for about 200,000 GPUs, and Colossus 2 targets gigawatt to multi-gigawatt capacity [15]; Meta has described multi-gigawatt campuses —Prometheus around 1 GW and Hyperion scaling toward 5 GW [14]. The International Energy Agency estimates that datacenters consumed about 415 TWh in 2024, roughly 1.5

The joint reading of Sections 4.1 and 6 thus defines three clearly separated power regimes: Latin American public science operates between 0.1 and 2 MW; the world’s frontier scientific supercomputers, in the tens of megawatts; and frontier AI training campuses, in the hundreds to thousands of megawatts. A credible first-generation Colombian national center belongs to the first regime —units-to-tens of petaflops and on the order of one megawatt, the scale of Santos Dumont—, with a plausible trajectory toward tens of megawatts only in later generations and under verified grid conditions. Framing the national ambition against Frontier or El Capitan therefore oversizes the cost and energy footprint of a realistic first step —precisely the calibration that public instruments have lacked. The category error of “AI factory” language does not consist in confusing adjacent regimes, but in skipping two at once: from the realistic sub-megawatt ambition to the regime of hundreds to thousands of megawatts, a jump of three to four orders of magnitude that turns a scientific-infrastructure debate into an energy-planning problem that the current grid and permitting framework are not designed to absorb.

7 National Grid Capacity and the Geography of Datacenter Feasibility

The preceding regimes acquire their real meaning only when placed against the capacity of the grid that would have to supply them. According to the system operator, Colombia’s National Interconnected System had an installed capacity of 21,315.61 MW in June 2026, against a maximum power demand of 12,475 MW [26] —a nominal reserve margin below 9 GW. The generation matrix remains hydro-dominated (about 62.8% hydro, 29.6% thermal, and 7.6% solar), which couples supply to hydroclimatic variability. The forward balance is tight rather than comfortable: UPME expansion tracking shows new firm capacity entering service well behind schedule, so that margin should be read as an upper bound [27]. This 2026 base —about 21 GW installed and 12.5 GW of peak demand— is the physical envelope any datacenter ambition must fit within.

Figure 3 places the three power regimes of Sections 4.1 and 6 on that envelope. A first-

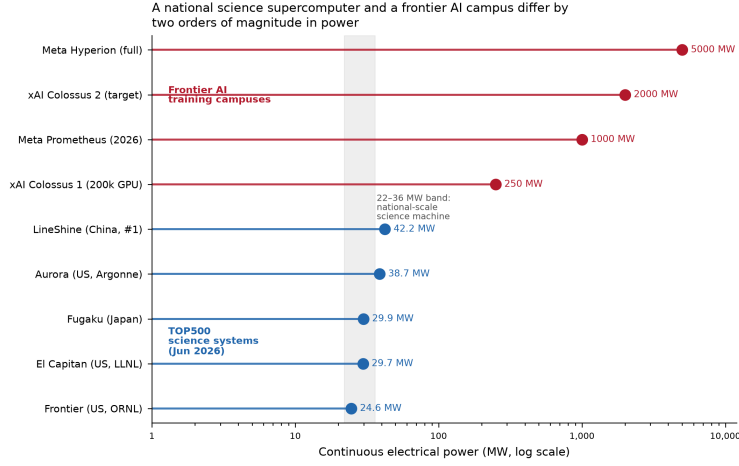


Figure 2: Continuous electrical power (logarithmic scale) of the five scientific supercomputers with the highest consumption in the June 2026 TOP500 list [16] (blue) versus reported and announced frontier AI training campuses (red). The shaded band marks the 22–36 MW range containing the world’s frontier scientific supercomputers; as Section 4.1 shows, Latin American public-science systems—the realistic entry regime for Colombia—operate one to two orders of magnitude below that band, and frontier AI campuses one to two orders above it.

generation national scientific machine on the order of 5 MW represents 0.02% of installed capacity; a frontier scientific supercomputer of 30 MW, about 0.14%. These loads are electrically negligible: they fit within existing urban infrastructure and do not alter national planning. A single frontier AI campus is a categorically different object. Colossus 1, at roughly 250 MW, already equals 1.2% of the *entire* national installed capacity; a gigawatt-class campus such as Meta’s Prometheus equals about 4.7%; and a 5 GW campus of the Hyperion class would equal 23.5% of everything Colombia has installed—roughly 40% of national peak demand, and about 2.3 times the peak load of the entire Bogotá metropolitan area. In a system already running a negative firm-energy margin, this is not an infrastructure detail to be resolved later but a national-planning impossibility on the current grid.

The constraint sharpens at the level of individual cities. Figure 4 compares the estimated metropolitan peak demand of Colombia’s four largest urban systems—Bogotá (Enel-Codensa area, on the order of 2,150 MW), Medellín (EPM, whose service territory covers 21.11% of national demand, on the order of 1,500 MW), Cali (Valle, on the order of 1,050 MW), and Barranquilla (Air-e, in the Caribbean region that is the country’s largest consumer at about 2,028 GWh per month, on the order of 900 MW)—against representative compute loads. The city figures are transparent order-of-magnitude estimates apportioned from the operator’s verified regional and utility-level anchors, not metered substation totals. Against them, a 5 MW national HPC is imperceptible ($\leq 0.6\%$ of any city’s peak) and a 30 MW scientific supercomputer remains modest (1.4% of Bogotá’s, 3.3% of Barranquilla). A 250 MW AI datacenter, by contrast, would draw 11.6% of Bogotá’s entire peak, 16.7% of Medellín’s, 23.8% of Cali’s, and 27.8% of Barranquilla’s—a single facility competing with a substantial fraction of a major city’s total electrical demand.

The decisive test is the intermediate city, where much of Colombia’s regional scientific capacity actually sits (SC3-UIS in Bucaramanga, BIOS in Manizales). Taking a representative intermediate-city peak of about 300 MW, the Santos Dumont use case is instructive: the LNCC’s national machine draws on the order of 1–3 MW, and even a 5 MW Colombian national HPC would consume only about 1.7% of such a city’s entire peak demand—comfortably absorbable. A 30 MW frontier scientific supercomputer would still be manageable at roughly 10% of the city’s peak, though it would already require deliberate coordination with the operator. But

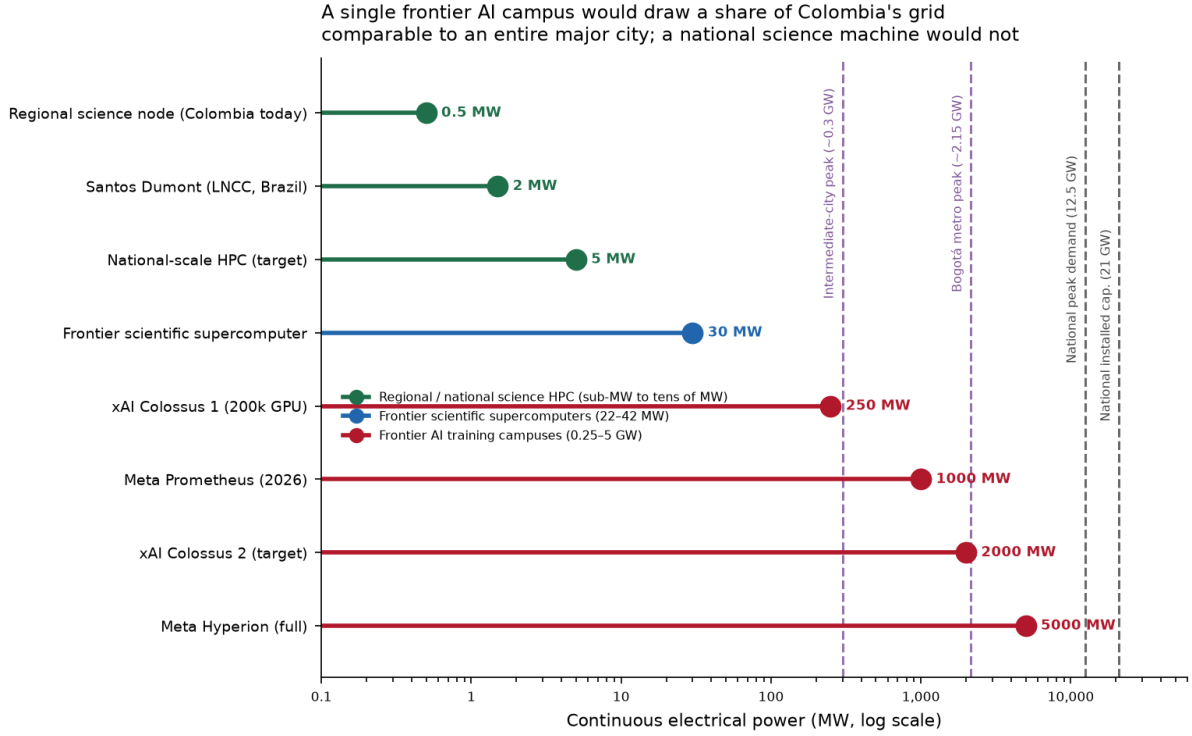


Figure 3: Continuous electrical power (logarithmic scale) of representative compute systems—regional and national scientific HPC (green), frontier scientific supercomputers (blue), and frontier AI training campuses (red)—against Colombian grid reference levels (dashed): an intermediate-city peak (~ 0.3 GW), the Bogotá metropolitan peak (~ 2.15 GW), national peak demand (12.5 GW), and national installed capacity (21 GW). A national science machine is electrically negligible; a single frontier AI campus consumes a share of the national grid comparable to an entire major city. Verified national figures from XM [26]; system loads from [16, 15, 14]; city and intermediate-city levels are author estimates derived from verified regional anchors.

a single 250 MW AI datacenter would demand about 83% of the intermediate city’s entire peak load: hosting it would mean nearly doubling the city’s firm generation and transmission, with dedicated substations and co-located generation, before a single accelerator is switched on. This is the concrete content of the “category error” diagnosed in Section 6: regional and national scientific HPC can be sited across the existing Colombian grid essentially anywhere; frontier AI campuses fit nowhere on that grid without being conceived, from the outset, as energy-infrastructure projects with their own generation. Feasibility, not ambition, is what the geography dictates.

8 Technological Sovereignty as Operational Capacity

Technological sovereignty is usually formulated in declarative terms, but for a scientific community its meaning must be operational. Sovereign capacity does not exclude international cooperation or the use of imported technology; it implies preserving the ability to inspect, validate, adapt, host, and sustain critical components without total dependence on external decisions.

In AI and HPC, that sovereignty can be disaggregated into at least four dimensions. Verification sovereignty: the capacity to contrast AI results against reliable scientific methods and one’s own testing infrastructure. Data sovereignty: jurisdictional and technical control over sensitive or strategic datasets. Infrastructure sovereignty: the ability to operate critical workloads without depending exclusively on the access, prices, or availability defined by foreign clouds. Talent

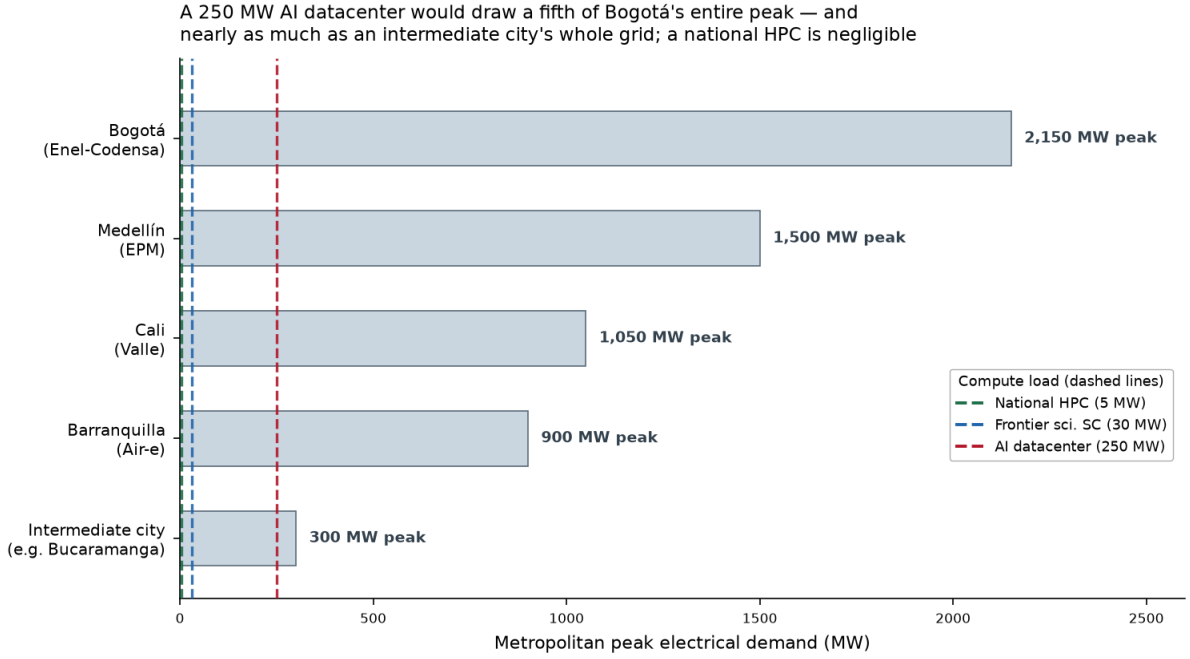


Figure 4: Estimated metropolitan peak electrical demand of Colombia’s four largest urban systems and a representative intermediate city, with three compute loads overlaid (dashed): a 5 MW national HPC, a 30 MW frontier scientific supercomputer, and a 250 MW AI datacenter. A national HPC is negligible everywhere; a 250 MW AI datacenter draws a fifth of Bogotá’s peak and about 83% of an intermediate city’s entire peak demand. City peaks are transparent order-of-magnitude estimates apportioned from XM regional and utility-level anchors [26]; they are not metered substation totals.

Table 3: The three Ministry of Science instruments analyzed in this article.

Instrument	Year	Axis	Role in this analysis
Call 46 (SGR)	2025	AI infrastructure	Design realism of a single national AI infrastructure project [6]
Call 966	2025	Quantum & AI	Results of the fundable bank analyzed as evidence [7, 9]
Call 976	2026	Quantum & AI (territories)	Extension of the same policy grammar to the territories [8]

sovereignty: local teams capable of administering systems, optimizing software, interpreting model behavior, and maintaining complex services.

Thus defined, sovereignty is not measured by the mere existence of hardware, but by the articulation among infrastructure, expert knowledge, technical governance, and institutional continuity. That is precisely the dimension in which Colombia faces a structural challenge today.

9 Lessons from Recent Public-Policy Instruments

To keep three similarly named instruments distinct, Table 3 associates each with its year, thematic axis, and role in this analysis.

Minciencias’ SGR Call 46, "Colombia Inteligente: Infrastructure for the Development of Artificial Intelligence" [6], illustrates the relationship between normative design and real infrastructure. It recognizes genuine gaps —low national compute capacity, weakness in datacenters and supercomputing, digital-talent shortages, and lack of secure processing and storage infrastructure— and mentions high-availability AI services, supercomputers, large-scale model training and deployment, high-density cabinets, energy efficiency, cooling, cybersecurity, and the Tier III standard.

The instrument nevertheless reveals a tension. Its territorial approach rewards projects executed outside Bogotá/Cundinamarca, Antioquia, Valle del Cauca, and Atlántico, with the legitimate purpose of decentralizing science. But in high-density AI infrastructure, siting is not merely a distributive variable: site selection depends on hard conditions —substation access, supply quality, cooling, backbone connectivity, logistics, seismic and climate risk, physical security, and qualified operators. When these appear only as desirable attributes rather than as central technical evaluation filters, the outcome can be inconsistent: a call discursively aligned with sovereignty and the democratization of knowledge, but without internalizing the material conditions that make it viable.

It should be acknowledged that this territorial bias is not an arbitrary decision by the instrument’s designer: SGR resources carry a constitutional and legal mandate of territorial distribution, so the call operates within constraints it cannot ignore. The critique, therefore, is directed not at the mandate but at how it is reconciled with technical feasibility: a mature policy can separate the siting filter for the high-density core —evaluated exclusively with hard criteria of site, energy, and operations— from the distributive criteria, applied to federated access, training, lower-density regional nodes, and the distribution of benefits, dimensions where decentralization is indeed viable and desirable. With that separation, the SGR territorial mandate and the physical viability of the infrastructure cease to be in conflict.

Its structure also deserves attention: a single national project, funded over 48 months with a considerable public purse. This favors coordination but concentrates execution risk and, if conceived as an isolated emblem, may reinforce a well-known regional practice: visible capital expenditure without equivalent strengthening of operations, interoperability, support, and

talent.

The lesson is not that the objective is wrong, but that public instruments for advanced infrastructure must be designed with the same realism they demand of the solutions they fund. Energy, cooling, continuity, technical siting, grid integration, and operational sustainability should operate as structural evaluation dimensions, not as secondary components of the institutional narrative.

10 Special Case: Colombia Inteligente 2026

The "Colombia Inteligente 2026" call (Call 976), addressed simultaneously to Artificial Intelligence and to Quantum Science and Technologies, shows another variant of the same problem. It seeks applied research, technological development, and innovation with tangible impact in the territories; it requires articulation with local communities, promotes social appropriation of knowledge, awards points for execution in PDET or ZOMAC municipalities, and proposes trajectories that at times aspire to levels near TRL 8 or TRL 9. That structure is reasonable for AI applications grounded in local data, sectoral pilots, and progressive adoption, but it becomes hard to sustain when transferred almost unchanged to the quantum field.

The problem is not territorial relevance: sublines such as quantum sensing for agriculture, health, environment, or demining do respond to regional needs. The difficulty is that the call groups very different instruments under a single grammar —algorithms, simulation, integrated and photonic circuits, ultra-secure communications, quantum internet, metrology, sensors, and materials. Some lines approach applied validation; others remain tied to laboratories, specialized infrastructure, scarce talent, dependence on external hardware, and maturation horizons incompatible with uniform requirements of community appropriation and rapid adoption.

The tension is greatest in quantum computing. Asking a project to respond to a specific territorial problem, involve at least 250 children, adolescents, and young people in programming and mathematics training, generate contextualized social appropriation, articulate communities, demonstrate benefits to end users, and simultaneously advance toward sustained operation superimposes logics that do not always converge. In applied AI that chain is difficult but intelligible; in quantum computing, value concentrates first in algorithmic experimentation, advanced training, validation against classical methods, and the construction of capacities prior to any broad adoption.

The asymmetry is inscribed in the very design of social appropriation: for AI, direct involvement of organizations is expected in problem formulation, pilots, and feedback; for quantum, the standard shifts toward information, consultation, and public dissemination, acknowledging that participation occurs at a greater distance from the technical core. The difference is reasonable, but it reveals that the instrument does not sufficiently distinguish between democratizing access to quantum knowledge and demanding territorial appropriation of a technology still far from direct community uses; conflating the two rewards narratives of premature impact over realistic capacity-building trajectories.

In public-policy terms, the call is right to open space for quantum technologies and to convene national experts, but it treats under a homogeneous framework technologies with deeply heterogeneous innovation economics. Quantum computing requires differentiated criteria: more weight on scientific solidity and technical validation than on immediate impact; explicit recognition of hardware and quantum-cloud dependencies; goals of advanced training and specialized software instead of early territorial adoption; and horizons compatible with a frontier technology whose utility can be strategic without yet being massive. Without that adjustment, quantum computing is asked to behave like a mature digital technology when its short-term national value lies in research capacity and sovereign positioning.

The results of the definitive bank of the Quantum Science and Technologies axis of Call 966 of 2025 reinforce this reading: of 27 registered projects, 19 met requirements, 18 were eligible, and only 6 were fundable [9] (Table 4).

Table 4: Selection funnel of the Quantum Science and Technologies axis of Call 966 of 2025. Source: definitive bank of eligible and fundable projects [9].

Stage	Projects
Registered	27
Met requirements	19
Eligible	18
Fundable	6

That reduction suggests friction between the volume of interest and the proposals’ capacity to satisfy the instrument. More revealing is that the fundable projects are dominated by sensing, monitoring, secure communications, energy grids, traceability, and agro-industrial applications: lines where the link between quantum technology and applied impact can be narrated more directly, with plausible pilots, users, and metrics. By contrast, quantum computing, the quantum internet, or advanced hardware, however valuable, can hardly demonstrate territorial impact within those timelines and formats.

Thus, the bank does not indicate failure, but the limits of the instrument. Under a strict criterion of territorial impact, only part of the fundable portfolio —those tied to sensing, monitoring, traceability, or applied security— has a defensible route to verifiable results; the rest may be interesting, but their connection to communities and local adoption appears more fragile or dependent on compliance narratives. The distance between what the policy says it wants and what certain frontier technologies can deliver remains unresolved. Since Call 976 of 2026 extends the same grammar of Call 966 to the territories, it is reasonable to anticipate —as inference, not observation— that it will reproduce a similar selection pattern as long as it does not incorporate criteria differentiated by technology type.

11 The National AI Policy and the Legislative Landscape

The instruments discussed operate within the framework set in 2025 by CONPES 4144, Colombia’s first National Artificial Intelligence Policy [20], which is illustrative precisely for the distance between what it names and what it funds. It is organized around six axes —ethics and governance, data and infrastructure, research and innovation, capacity development, risk management, and adoption— and articulates 106 actions with an indicative budget on the order of COP 479 billion through 2030. The diagnosis acknowledges that the country lacks high-performance computing and large-scale datacenter capacity.

The discontinuity arises when that diagnosis is translated into prioritized investment. The data-and-infrastructure actions are oriented toward data availability, connectivity, interoperability, and edge or cloud services, not toward the compute substrate that —as argued in Sections 5 and 6— serious adaptation and deployment of foundation models require, not even at the entry regime of Section 4.1. The policy funds data sovereignty and part of infrastructure sovereignty at the connectivity layer, but leaves the compute-and-energy substrate —the binding constraint— to generic references and future instruments. Diagnosing the absence of supercomputing without funding it shifts the problem to sectoral calls that are too small to close it.

The legislative record reinforces this. In the 2024–2025 legislatures, about ten AI-related bills were filed; nine were shelved and only one became law: a Criminal Code reform sanctioning identity impersonation through AI-generated content [21]. The surviving initiatives —including the unified bill PL 043-2025, which names "technological sovereignty" among its principles [29]— deal with governance, ethics, fundamental rights, transparency, and fraud, not with the material base of compute and energy; even the IDEC bill on the State’s digital infrastructure addresses data and interoperability, not computational or energy capacity [30]. Table 5 summarizes the most prominent initiatives.

Table 5: Orientation of the most prominent Colombian legislative initiatives on AI (2024–2025). Here "infrastructure" designates the compute-and-energy substrate, distinct from data/interoperability.

Initiative	Main orientation	Status
Criminal Code reform (AI impersonation) [21]	Fraud / criminal law	Enacted
PL 043-2025 (unified AI bill) [29]	Governance, ethics, rights; names "technological sovereignty" as a principle	In progress
IDEC bill (State digital infrastructure) [30]	Data and interoperability	In progress
Other AI bills (2024–2025)	Ethics, proper use, transparency, risk	Shelved

Taken together, CONPES 4144 and the legislative landscape describe a policy apparatus that has learned to name sovereignty and infrastructure, but has not yet committed to funding the compute-and-energy substrate on which operational AI capacity depends. This is not a rhetorical failure, but one of budgeting and sequencing, exactly the kind that this work’s operational definition of sovereignty seeks to expose.

12 Strategic Implications for Colombia

The evidence reviewed allows a more coherent agenda for the national baseline. First, Colombia should prioritize the integration of its existing ecosystem before hyperscale expansion: a stable coordination mechanism among universities, CyberColombia [3], RENATA, SCALAC [1], RedCLARA [19], and public entities would build interoperability, standards, and a verifiable roadmap. Second, its first major bet should be a national advanced computing center with clear scientific functions —simulation, validation, sensitive inference, adaptation of open models, and hybrid HPC–AI workflows in priority sectors— sized at the entry regime identified in Section 4.1 (tens of petaflops, on the order of one megawatt, with a verifiable growth path), anchoring and elevating the capacity of the existing network without replacing it. Third, talent development must become explicit policy: national survey data show sustained training demand and institutional gaps; without system administrators, systems engineers, network specialists, numerical-analysis experts, and advanced scientific users, new infrastructure will reproduce underutilization. Fourth, decentralization is desirable in access, training, services, and collaboration, but high-density physical infrastructure must be sited according to technical feasibility: regional nodes, federated access, robust academic networks, and a core located where operations are truly sustainable, a scheme that also honors the SGR territorial mandate without compromising physical viability (Section 8).

13 Conclusion

The convergence between HPC and AI is not the substitution of scientific computing by opaque model outputs, but a reorganization of computational practice in which acceleration acquires value only when accompanied by verification, reproducibility, and institutional control.

The three Minciencias instruments examined share a design problem: they recognize real needs, but translate them into instruments lacking the technical rigor needed to reconcile objectives, material constraints, and evaluation criteria. In Call 46, the ambition to build advanced AI infrastructure coexists with territorial and distributive requirements that can collide with

hard demands of energy, site, grid, cooling, and continuous operation—a manageable tension if the technical filter for the core is separated from the distributive criteria of access, but not if the two are conflated. In Calls 966 and 976, the logic of territorial impact, social appropriation, mass training, and technological maturation reaches lines of quantum science and quantum computing whose trajectory does not respond to the same timelines, infrastructures, or forms of adoption; the fundable bank of Call 966 shows that the system more easily selects what translates into immediate applications, leaving strict frontier bets in an awkward position. In all three cases, the instrument is too demanding in peripheral dimensions and too imprecise in the central technical conditions.

Two empirical findings frame that diagnosis. The first is the scale calibration: Latin American public science operates in the sub-megawatt to few-megawatt regime—Santos Dumont delivers 14.3 PFlop/s at 0.31 MW—, two orders of magnitude below the world scientific frontier and up to four below frontier AI training campuses. A credible first Colombian national center belongs to that entry regime, and the instruments analyzed lack precisely that calibration: they speak the language of the "AI factory" without sizing the realistic first step. The second is the gap between naming and funding: CONPES 4144 diagnoses the absence of supercomputing and the legislature invokes technological sovereignty, but neither the policy nor the bills commit budget to the compute-and-energy substrate. Colombian AI policy discursively oversizes what it materially underfunds, and that double distortion—frontier-level rhetorical ambition, effective investment below even the entry regime—is the central finding of this work.

The resulting risk runs deeper than administration: by accumulating incompatible requirements, proponents optimize for eligibility rather than scientific relevance. The design rewards proposals that simultaneously promise immediate territorial impact, broad appropriation, extensive training, sustainability, novelty, and feasibility, even when the technology does not allow it, biasing selection toward projects that learn the instrument's language while diluting their technical ambition.

The underlying problem is one of institutional architecture of technical judgment. Technological sovereignty requires instruments capable of discriminating among technology types, maturation horizons, and legitimate forms of impact, accepting that some bets demand heavy infrastructure and physical feasibility, while others require advanced training, experimental validation, and the patient construction of capacities. Without that differentiation, calls simultaneously demand immediate social pertinence and serious frontier development, with no framework to reconcile them. The challenge for Colombia is not only to invest more or broaden the discourse on AI and quantum technologies, but to design instruments with greater technical discipline, calibrated at the right scale, with less friction among requirements and greater capacity to attract genuinely relevant projects rather than projects adapted to survive the form.

Declaration on Affiliation and Conflicts of Interest

The author is affiliated with CyberColombia, an organization that has coordinated HPC training and research activities in Colombia for more than a decade—including nine editions of the HPC Summer School—and that produced the reports [3, 4, 5] used here. This position provides direct access to the national survey data, but it implies that the author analyzes evidence from his own organization; to mitigate that risk, claims based on CyberColombia sources are triangulated, where possible, with independent sources (SCALAC, TOP500, CONPES, ILIA), and where no independent corroboration exists, the text says so explicitly.

References

- [1] SCALAC HPC Observatory: *Expanding on Latin American and Caribbean HPC Infrastructure Analysis*, version 3.1.0 (2025). https://scalac.redclara.net/images/2025/SCALAC_

HPC_Infra_3_1_0.pdf (accessed June 30, 2026).

- [2] CyberColombia: *Estado del HPC en Colombia — Respuestas de la encuesta (2025)*. Open dataset (survey instrument and aggregated institutional responses, March–October 2025). Zenodo, 2026. <https://doi.org/10.5281/zenodo.21209682> (DOI: 10.5281/zenodo.21209682, accessed June 2026).
- [3] CyberColombia: *Driving HPC and AI-powered solutions for Colombia’s digital future and Latin America*. <https://cybercolombia.org/> (accessed June 23, 2026).
- [4] CyberColombia: *Status of HPC in Colombia 2026*. https://www.cybercolombia.org/assets/docs/Status_of_HPC_in_Colombia_2026.pdf (accessed June 25, 2026).
- [5] CyberColombia: *Introduction to Supercomputing*. Technical report (2025).
- [6] Ministry of Science, Technology and Innovation of Colombia: *Terms of Reference, SGR Call 46, Colombia Inteligente: Infrastructure for the Development of Artificial Intelligence* (2025). <https://minciencias.gov.co/convocatorias/plan-convocatorias-asctei-2025-2026/convocatoria-colombia-inteligente-infraestructura> (accessed June 2026).
- [7] Ministry of Science, Technology and Innovation of Colombia: *Call No. 966 of 2025, Colombia Inteligente: Quantum Science and Technologies and Artificial Intelligence*. <https://minciencias.gov.co/convocatorias/convocatoria-colombia-inteligente-ciencia-y-tecnologias-cuanticas-e-inteligencia> (accessed June 30, 2026).
- [8] Ministry of Science, Technology and Innovation of Colombia: *Call No. 976 of 2026, Colombia Inteligente: Quantum Science and Technologies and Artificial Intelligence for the Territories*. <https://minciencias.gov.co/convocatorias/convocatoria-colombia-inteligente-2026> (accessed June 30, 2026).
- [9] Ministry of Science, Technology and Innovation of Colombia: *Definitive bank of eligible and fundable projects, Quantum Science and Technologies axis, Call No. 966 of 2025* (published September 16, 2025). https://minciencias.gov.co/sites/default/files/publicacion_del_banco_de_proyectos_elegibles_y_financiables_-_eje_tematico_inteligencia_artificial_de_la_convocatoria_966-2025_0.pdf (accessed July 2026).
- [10] Meta Engineering: *Introducing the AI Research SuperCluster — Meta’s AI supercomputer* (2022). <https://engineering.fb.com/2022/01/24/ai-research/ai-rsc/>
- [11] NVIDIA: *NVIDIA DGX H200*. <https://www.nvidia.com/en-us/data-center/dgx-h200/> (accessed June 23, 2026).
- [12] Jouppi, N.P., Lakshmanamurthy, S., Young, C., Patterson, D.: *Google’s training supercomputers from TPU v2 to Ironwood: architectural stability, scale, resilience, power efficiency, and sustainability across five generations*. arXiv:2606.15870 (2026).
- [13] International Energy Agency: *Energy and AI* (2025). <https://www.iea.org/reports/energy-and-ai> (accessed June 30, 2026).
- [14] Meta Platforms: public announcement by M. Zuckerberg on the Prometheus (1 GW) and Hyperion (up to 5 GW) superclusters, July 2025; complementary reporting by TechCrunch and Data Center Frontier. <https://techcrunch.com/2025/07/14/mark-zuckerberg-says-meta-is-building-a-5gw-ai-data-center/> (accessed June 30, 2026).

- [15] Technical coverage of xAI’s Colossus campus in Memphis (Colossus 1, 200,000 GPUs and estimated demand on the order of 250 MW; Colossus 2 expansion at gigawatt scale), 2024–2026: Data Center Dynamics and SemiAnalysis; public documentation from Memphis Light, Gas and Water on supply requests. <https://newsletter.semianalysis.com/p/xais-colossus-2-first-gigawatt-datacenter> (SemiAnalysis; accessed July 2026).
- [16] TOP500: *TOP500 List — June 2026*. <https://www.top500.org/lists/top500/2026/06/> (accessed June 30, 2026).
- [17] Li, B., Roy, R.B., Wang, D., Samsi, S., Gadepally, V., Tiwari, D.: *Toward sustainable HPC: carbon footprint estimation and environmental implications of HPC systems*. In: SC23: International Conference for High Performance Computing, Networking, Storage and Analysis. IEEE (2023).
- [18] CENIA: *Latin American Index of Artificial Intelligence*. <https://indicelatam.cl/> (accessed June 23, 2026).
- [19] RedCLARA: *Advanced Academic Network Services for Latin America*. <https://www.redclara.net/> (accessed June 23, 2026).
- [20] National Council for Economic and Social Policy (CONPES): *CONPES Document 4144: National Artificial Intelligence Policy*. National Planning Department (DNP), Bogotá, Colombia, 2025. <https://colaboracion.dnp.gov.co/CDT/Conpes/Economicos/4144.pdf>
- [21] Congress of the Republic of Colombia: *Law 2502 of 2025* (enacted July 28, 2025), which adds an aggravating circumstance to Article 296 of Law 599 of 2000 (Colombian Criminal Code) to sanction identity impersonation (*falsedad personal*) committed by means of artificial intelligence. Originated as Bill 225 of 2024 Senate. Bogotá, Colombia.
- [22] National Laboratory for High Performance Computing (NLHPC), Universidad de Chile / Center for Mathematical Modeling: *Infrastructure: Guacolda-Leftraru Epu supercomputer*. Santiago, Chile, 2026. <https://www.nlhpc.cl/infraestructura/>
- [23] National Supercomputing Laboratory of Southeast Mexico (LNS), Benemérita Universidad Autónoma de Puebla: *Supercomputing infrastructure (Cuetlaxcoapan)*. Puebla, Mexico, 2026. <https://lns.buap.mx/>
- [24] Laboratório Nacional de Computação Científica (LNCC): *2025 upgrade of the Santos Dumont supercomputer*. Petrópolis, Brazil, 2025.
- [25] RISC2 Project: *A network for supporting the coordination of High-Performance Computing research between Europe and Latin America*. European Commission, H2020 grant agreement no. 101016478, 2021–2023. CORDIS: <https://cordis.europa.eu/project/id/101016478>
- [26] XM S.A. E.S.P. (operator of the National Interconnected System): reports on net effective capacity and the composition of the SIN generation matrix. Medellín, Colombia. <https://www.xm.com.co/>
- [27] Mining and Energy Planning Unit (UPME): *Reference Expansion Plan, Generation–Transmission* (current version). Bogotá, Colombia. <https://www.upme.gov.co/>
- [28] Enerdata: *Colombia Energy Information / World Energy & Climate Statistics — Yearbook*. Grenoble, France, 2025. Electricity consumption per capita $\approx 1,600$ kWh/yr (2024). <https://www.enerdata.net/estore/energy-market/colombia/>

- [29] Congress of the Republic of Colombia: *Bill 043 of 2025 Senate / 324 of 2025 House (unified artificial intelligence bill)*. Filed in the Senate on July 28, 2025 by the National Government. First-debate report published in Gaceta del Congreso No. 2348 and No. 2349 of 2025 (errata in Gaceta No. 84 of 2026). Bogotá, Colombia.
- [30] Congress of the Republic of Colombia: *Bill 447 of 2024 House / 248 of 2024 Senate — Data Infrastructure of the Colombian State (IDEC) and interoperability of public information systems (“Data Law”)*. Filed in the House of Representatives on May 22, 2024 by the Ministry of ICT; approved in first debate in the Sixth Committee of the House on June 3, 2024. Bogotá, Colombia.